

BOOKEXCHANGE

Connecting Students. Saving Money. Reusing Books.

ENSE 281 | Scrum 1 Presentation| March 4, 2026

Meet the Team

Thomas P.

GitHub Administrator
Frontend Developer

Chop Peter

Project Manager

David

Lead Developer

Shaikh

QA Lead
Documentation

The Problem

Students spend **\$1,200+/year** on textbooks

Campus bookstore buyback: only **10-20%** of original price

No campus-specific platform exists to connect student buyers and sellers

Result:

- Buyers pay full price
- Sellers get almost nothing back
- Books sit unused or are discarded

Our Solution

A web application where students buy and sell used textbooks directly to each other



**Buyers save
30-70%**

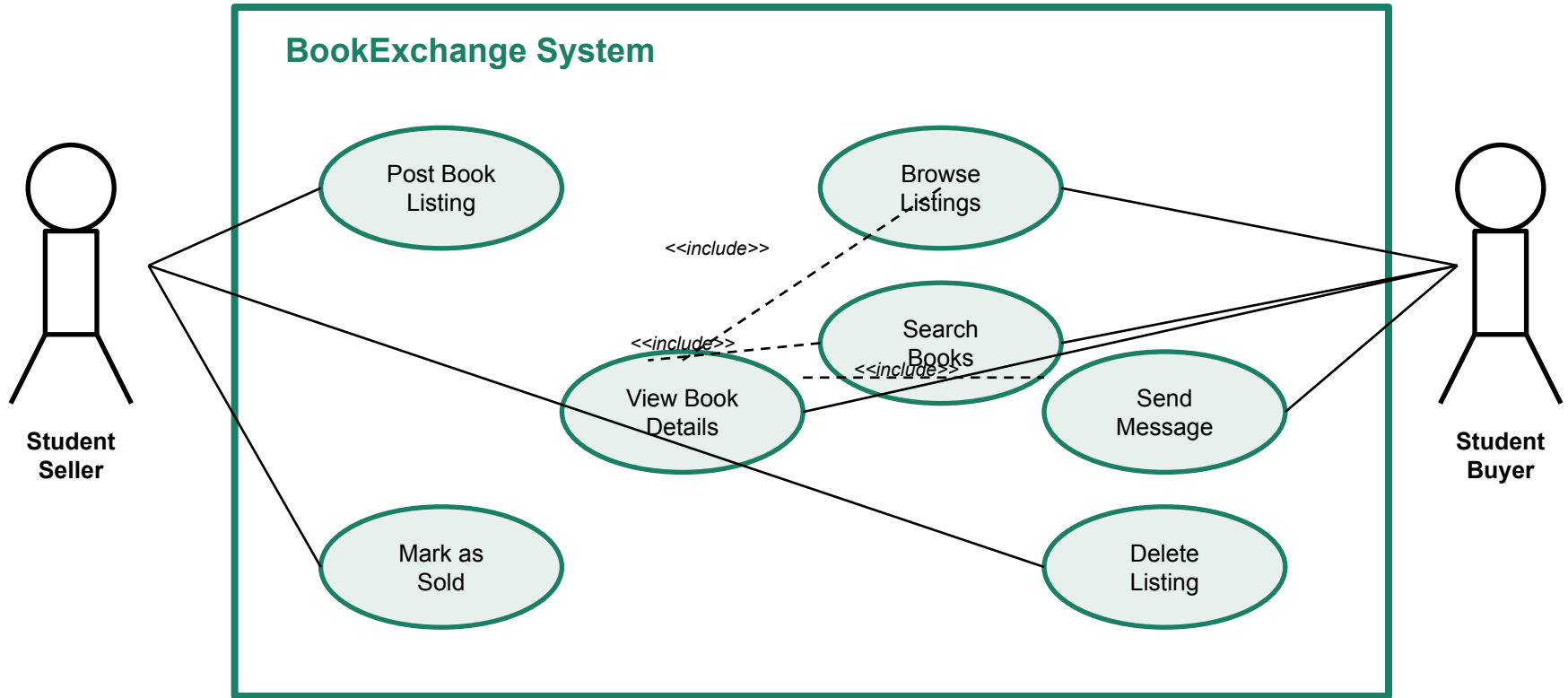


**Sellers get
fair value**

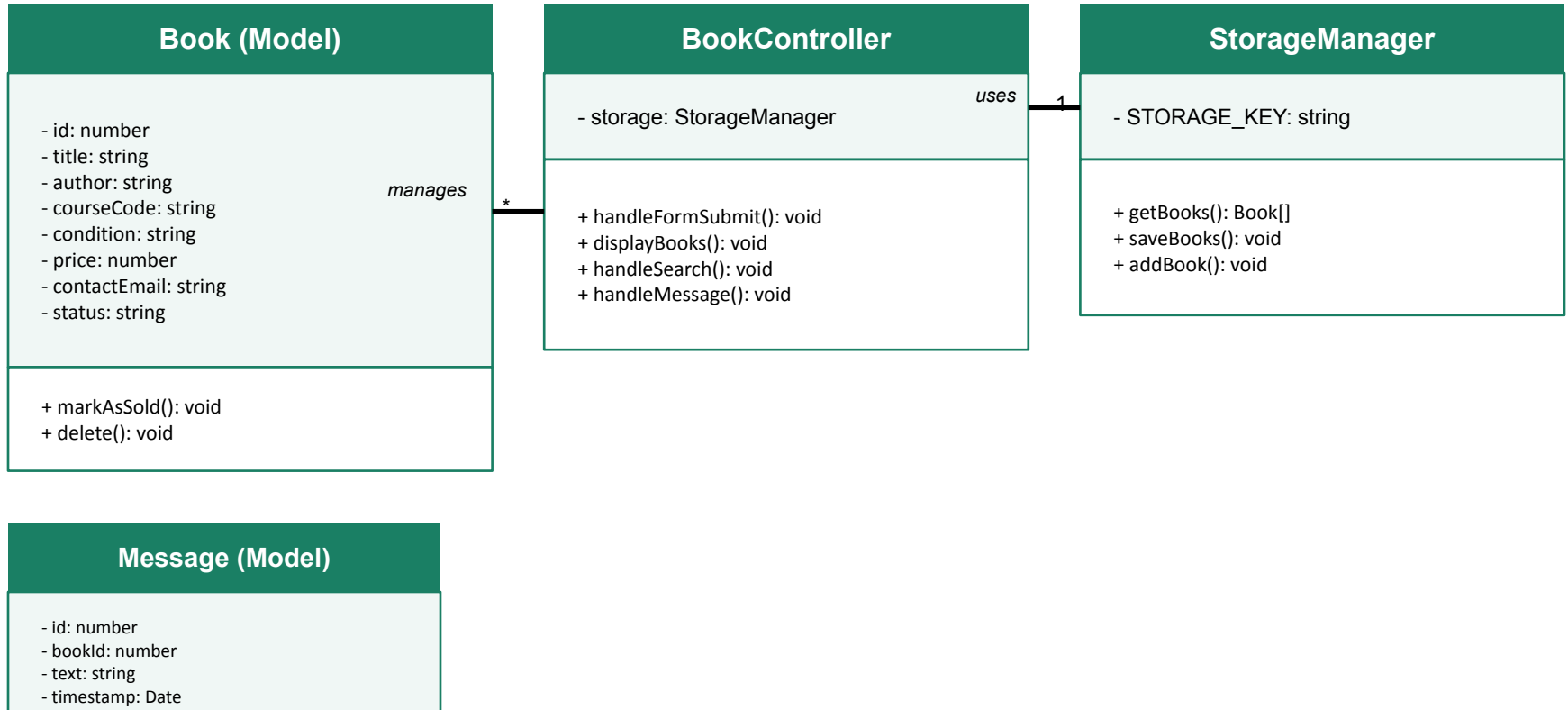


**Books get
reused**

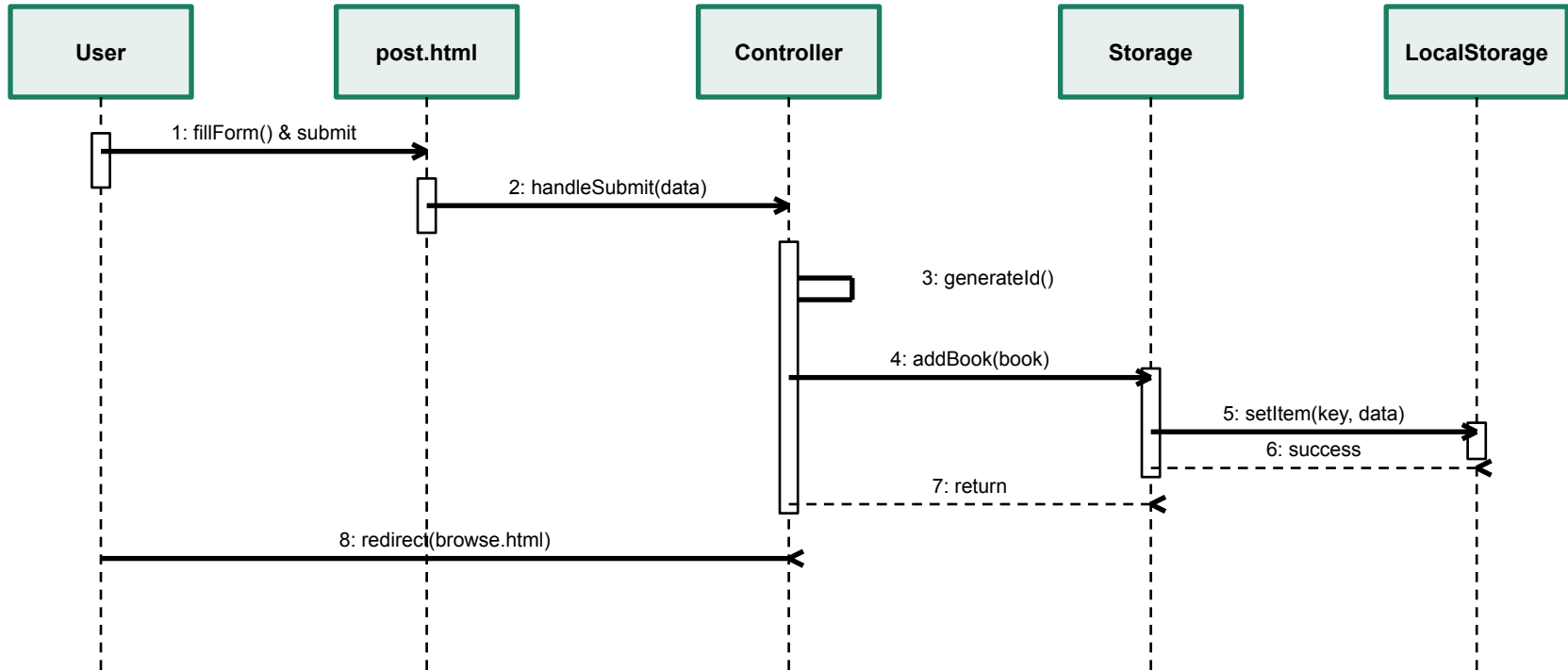
UML: Use Case Diagram



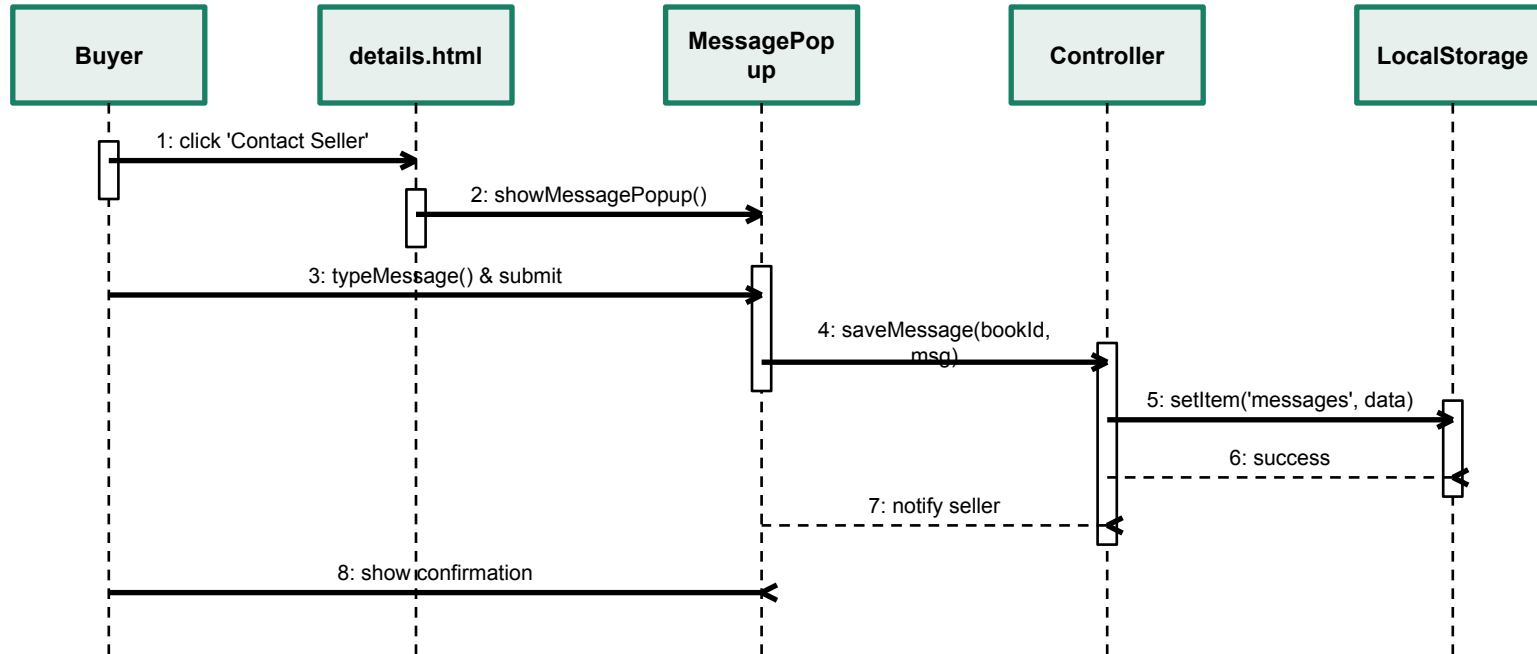
UML: Class Diagram (MVC Pattern)



UML: Sequence Diagram - Post Book



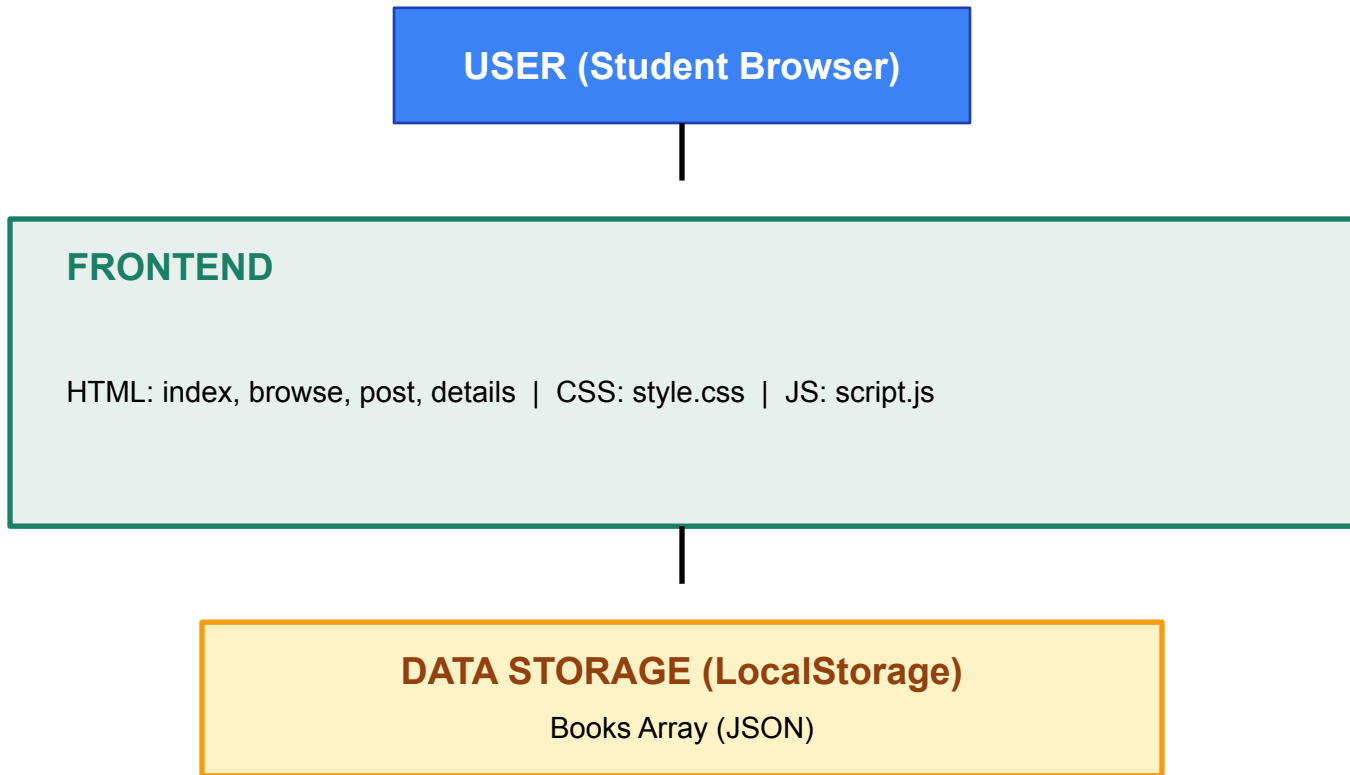
UML: Sequence Diagram - Send Message



Message stored temporarily. Seller gets notification badge. Both can share contact info privately in messages.

System Architecture

High-Level Design



Technology Stack

HTML5

Page structure

CSS3

Responsive styling

JavaScript

Vanilla JS (no frameworks)

LocalStorage

Browser-based data

Git/GitHub

Version control

VS Code

Development environment

Key Features (MVP)

1. Post Book Listing

Form with title, author, course code, condition, price, contact

2. Browse Listings

Grid display of all available books

3. Search Books

Filter by course code or title (case-insensitive)

4. View Details

Complete book info + seller contact

5. Mark as Sold

Remove from browse after transaction

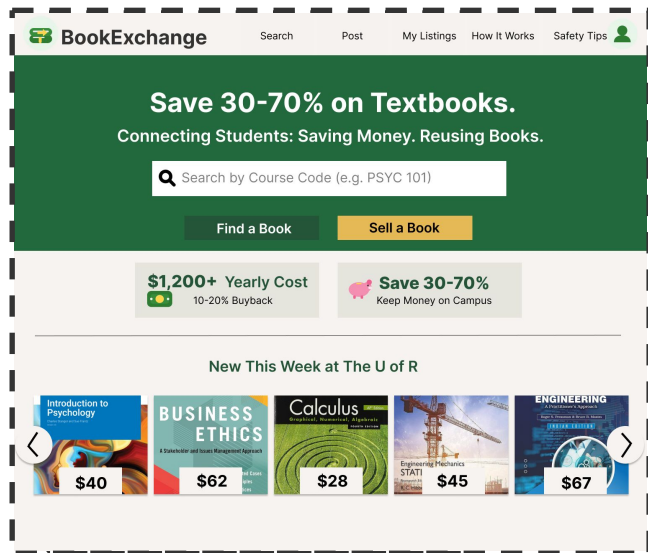
6. Delete Listing

Permanently remove unwanted listings

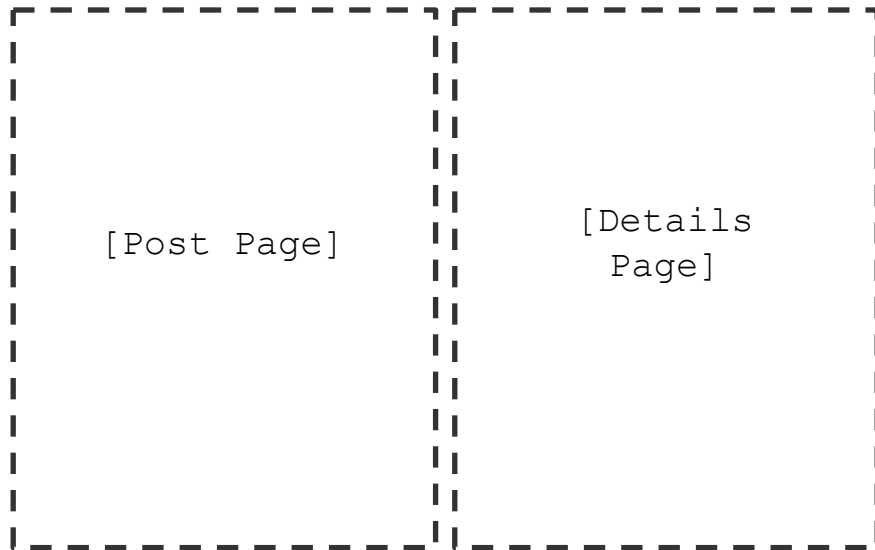
All processing happens client-side. No backend or server required.

UI Design

Low-Fidelity Wireframes



Welcome + navigation



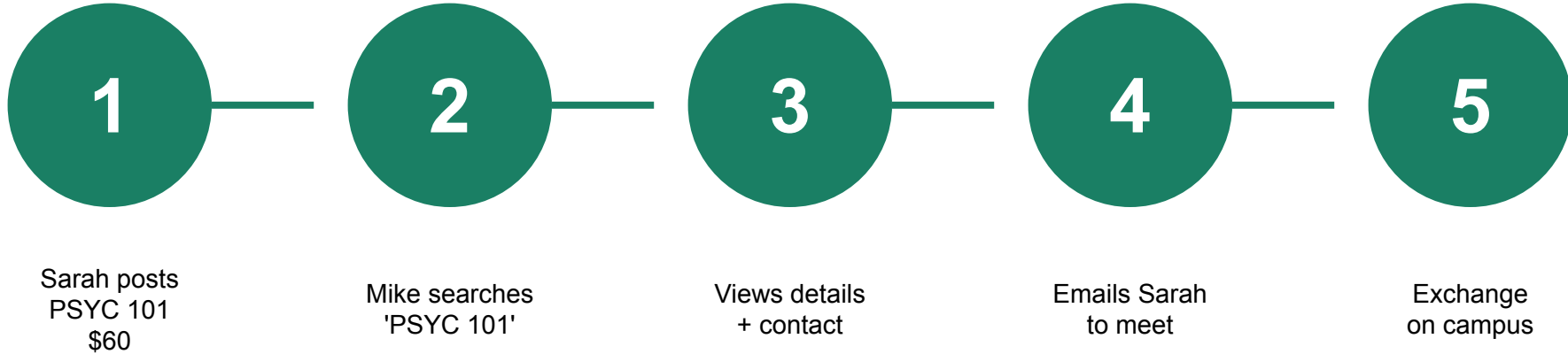
Form to add listing

Full book info

Note: Mockups designed for desktop and mobile responsiveness

User Flow Example

Scenario: Student buys a PSYC 101 textbook



Result: Thomas saves \$90, Sam gets \$60 (vs \$25 bookstore buyback). Both win!

Challenges & Solutions

Challenge: Privacy Concerns

Displaying seller email/phone publicly raises safety issues

Solution: Considering 'Reveal on Click' for MVP, in-app messaging for future

Challenge: LocalStorage Limitations

Data is browser-specific and can be cleared

Solution: For MVP, accept this trade-off. Future: migrate to Firebase

Challenge: First Web Development Project

Learning curve for team members new to web dev

Solution: Pair programming, regular check-ins, detailed documentation

Self-Reflection

How did you feel about this activity?

It was challenging but rewarding. We enjoyed designing a solution to a real student problem. The scope was manageable, which kept the project from feeling overwhelming.

What is your team most proud of?

Creating a complete system architecture from scratch and organizing our GitHub repository properly. We're also proud of addressing Tim's feedback about privacy concerns thoughtfully.

"We learned that good documentation and planning upfront saves time later in development."

Self-Reflection (Continued)

What did you learn about yourself?

We learned the importance of clear communication and task division. Using the RACI chart helped us understand our roles. We also learned to be comfortable asking for help when stuck.

How will you use what you learned going forward?

We'll continue using Git best practices, proper documentation, and systematic planning. The experience of building something from requirements to architecture will help in future projects.

What would you like more help with?

Best practices for handling user privacy and data security. Also guidance on when to add features vs keeping scope minimal for MVP.

Next Steps

Immediate (This Week)

- • Implement HTML/CSS for all 4 pages
- • Begin JavaScript functionality
- • Create UI mockups in Figma

Short-term (Next 2 Weeks)

- • Complete all 6 core features
- • Test with sample data
- • Address privacy concerns

Future Enhancements

- • Migrate to Firebase for real database
- • Add user accounts and authentication
- • Implement in-app messaging

Feedback

We would like to get YOUR feedback on our current website design. Please check out our prototype and fill out this google form. —>

Please

Prototype:



Google Form:



Thank You!

Questions?

GitHub: github.com/thomaskeithpersson-afk/281-Group-Project